FORCe window boundaries and filtering in movement-locked EEG

A matched two-participant pilot research note

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Abstract

Independent one-second FORCe calls can introduce discontinuities when reconstructed windows are joined. We evaluate those boundaries and an offline adaptation: continuous high-pass filtering, one-second cleaning, and low-pass filtering of a three-second segment before cropping. Four conditions separate cleaning and filtering effects. Across 2,562 matched trials, the +0.5 s Cz median jump was 3.778 μV with bandpass-first FORCe and 0.440 μV with split filtering, versus 0.414 μV in the split control. Finite-segment filtering residuals were below 10^{-9} μV, but broader cleaning-related differences remained. This pilot is descriptive; lower boundary steps or changed ERP waveforms do not establish preservation of underlying neural signals or better artifact removal.

1 Questions and scope

The primary question is whether window joins introduce unusually large changes and whether proposed processing reduces them without leaving broader transitions. The second question concerns short-segment filtering effects within the retained interval. FORCe is optional; practical consequences are evaluated rather than a benefit assumed. FORCe combines wavelets, independent-component analysis and thresholding [1]. Empirically calibrated settings are retained. The supplied Graz study discusses responses around 0.4–0.8 s [2]; the proposed join at +0.5 s intersects that interval.

2 Methods

2.1 Data and four processing conditions

Both pilot participants (mle01 and spe30), all eligible task recordings, and pre/post/retest sessions are included. Eight-channel EEG is sampled at 500 Hz; Cz and FCz are primary. Movement-locked extraction is $[-1.5, 1.5)$ s. The three cleaning windows are $[-1.5, -0.5)$, $[-0.5, 0.5)$ and $[0.5, 1.5)$ s: movement onset is no longer a join. Continuous filters are applied separately to valid recording segments. Acquisition-loss markers, nonfinite samples and timestamp gaps create boundaries. The complete three-second interval must lie outside two-second guards at each segment endpoint. This is an existing conservative policy, not a proven settling bound. Butterworth filters have order four and are applied forward/backward. The upper cutoff remains 20 Hz. The historical bandpass and separate high-/low-pass designs differ, so this evaluates practical pipelines rather than filter order alone.

ID	Continuous filtering	Three-second processing
1	1–20 Hz bandpass	No FORCe
2	1–20 Hz bandpass	Three 1 s FORCe calls
3	1 Hz high-pass	Three 1 s FORCe calls; 20 Hz low-pass
4	1 Hz high-pass	No FORCe; 20 Hz low-pass

All conditions are cropped to $[-1, 1)$ s, baseline-corrected over $[-0.2, 0)$ s, and rejected if absolute amplitude exceeds 100 μV. FORCe failures reject the complete trial; the existing positive 200 μV channel threshold excludes the unresolved interpolation branch. Source decomposition depth remains two, and no thresholds or library behaviour are changed. Condition 4 is a diagnostic control. Split/rejoin without cleaning is checked for exact sample preservation. Condition 3 is saved before and after low-pass filtering. Main comparisons use trials accepted in all four conditions; separate retention counts document excluded trials.

2.2 Boundary measurements

For trial i, channel c, and boundary b,

$$J_{icb} = |x_{ic}(b) - x_{ic}(b - 2 \text{ ms})|.$$

The same measurement at onset ($b = 0$) supplies a non-join location. Ordinary changes are absolute adjacent-sample differences in the retained epoch, excluding ± 50 ms around -0.5, 0 and $+0.5$ s. Each jump's empirical percentile and the fraction at or above the 99th percentile describe rarity, not an artifact label or formal significance. Percentiles are condition-specific; low-pass filtering changes the reference distribution. Define the fixed scale from condition 1:

$$S_{ic} = Q_{95}(x_{ic}^{(1)}) - Q_5(x_{ic}^{(1)}), \qquad J_{icb}^{*} = J_{icb}/S_{ic}.$$

This imposes no Gaussian assumption. Degenerate scales are not interpreted. Cz and FCz are primary; each trial's maximum-channel step is supporting information. A maximum is not assembled into a fictitious trial waveform.

2.3 Residual checks

A residual is a processing difference, not necessarily an error.

1. **Finite-segment filtering:** condition 4 minus continuous high-pass followed by continuous low-pass, after identical crop and baseline. This isolates applying the low-pass to three seconds instead of continuously.
2. **Injected boundary:** a unit persistent step or a one-sample spike is injected at each join into an otherwise unchanged no-FORCe sequence. By linearity, subtracting injected/uninjected outputs equals filtering the disturbance. Normalisation is by injected amplitude. Short- and long-segment responses use the actual low-pass coefficients and endpoint handling. Synthetic plots are before baseline correction to isolate filtering; real-trial residuals include baseline correction.
3. **Actual cleaning:** condition 3 minus 4, and condition 2 minus 1. Both include intended artifact removal and potentially unwanted changes; neither estimates neural error.

For real trials, $r_{ic}(t) = [x_A(t) - x_B(t)]/S_{ic}$. Median, 95th percentile and maximum absolute residual are measured in the full epoch, its outer 200 ms at each end, the 0.2–0.9 s response interval, and ± 50 ms at both joins. Values are reported in μV and reference-range fractions. No arbitrary cutoff declares a boundary resolved. A centring audit uses the first three eligible trials per recording. The existing wavelet/SOBI path is reconstructed without rejection or coefficient thresholding. Mean shifts and nonconstant RMS residuals distinguish level changes from broader reconstruction differences; the supplied implementation is not edited. For each participant/session and channel, displaced-minus-non-displaced averages use identical trials. Extrema and largest-absolute-peak times within 0.2–0.9 s describe waveforms, not validated component latencies.

2.4 Statistical interpretation

Two-participant results are descriptive. Time samples are not independent replicates, and trial counts cannot substitute for participant replication. No population-level significance or equivalence claim is made. Filtering may spread abrupt changes [3]; cropping cannot remove a join within the retained interval.

3 Results I: FORCe boundaries

There are 29 recordings, 2,915 trial rows, 2,799 eligible complete intervals, and 2,562 trials accepted by all four conditions.

Participant	Eligible	1	2	3	4	Common
mle01	1411	1179	1201	1203	1182	1176
spe30	1388	1386	1387	1387	1386	1386
Pooled	2799	2565	2588	2590	2568	2562

3.1 Magnitude and rarity

Pooled matched-trial statistics are shown below. J/S is expressed as a percentage. Rare means the fraction of jumps at or above each trial's 99th-percentile ordinary change. Participant-specific summaries appear in the appendix.

Ch.	Join	ID	Median (μV)	J/S (%)	Q_{95} (μV)	Rare (%)
Cz	-0.5	1	0.330	1.27	1.190	0.6
Cz	-0.5	2	1.550	5.82	8.976	42.3
Cz	-0.5	3	0.306	1.20	1.035	1.4
Cz	-0.5	4	0.327	1.25	1.182	0.6
Cz	+0.5	1	0.416	1.57	1.459	2.1
Cz	+0.5	2	3.778	14.17	16.329	66.9
Cz	+0.5	3	0.440	1.72	1.445	5.8
Cz	+0.5	4	0.414	1.56	1.452	2.2
FCz	-0.5	1	0.335	1.22	1.217	0.7
FCz	-0.5	2	1.689	6.08	10.294	43.7
FCz	-0.5	3	0.312	1.13	1.053	1.4
FCz	-0.5	4	0.334	1.20	1.200	0.7
FCz	+0.5	1	0.426	1.53	1.597	2.2
FCz	+0.5	2	4.205	15.14	18.369	68.7
FCz	+0.5	3	0.473	1.70	1.554	6.8
FCz	+0.5	4	0.423	1.53	1.589	2.2

At +0.5 s in Cz, median jumps are 3.778 μV for bandpass-first FORCe and 0.440 μV for split FORCe; the split control is 0.414 μV. Rarity fractions are 66.9% and 5.8% respectively. These are pooled pilot descriptions.

3.2 Paired boundary effects

Contrast A: condition 2 minus 1 (original FORCe effect). B: condition 3 minus 4 (proposed FORCe effect). C: condition 2 minus 3 (change in FORCe usage). A positive value means a larger jump in the first condition. These are medians of paired differences, not differences between marginal medians.

Ch.	Join	Contrast	Median ΔJ (μV)	$\Delta J/S$ (%)	Positive (%)
Cz	-0.5	A	1.099	4.25	85.1
Cz	-0.5	B	-0.027	-0.12	42.8
Cz	-0.5	C	1.156	4.48	87.0
Cz	+0.5	A	3.215	12.37	91.5
Cz	+0.5	B	0.015	0.06	51.7
Cz	+0.5	C	3.281	12.44	91.4
FCz	-0.5	A	1.285	4.51	85.6
FCz	-0.5	B	-0.026	-0.10	43.8
FCz	-0.5	C	1.307	4.76	87.9
FCz	+0.5	A	3.693	13.14	91.5
FCz	+0.5	B	0.020	0.07	52.5
FCz	+0.5	C	3.683	13.13	92.1

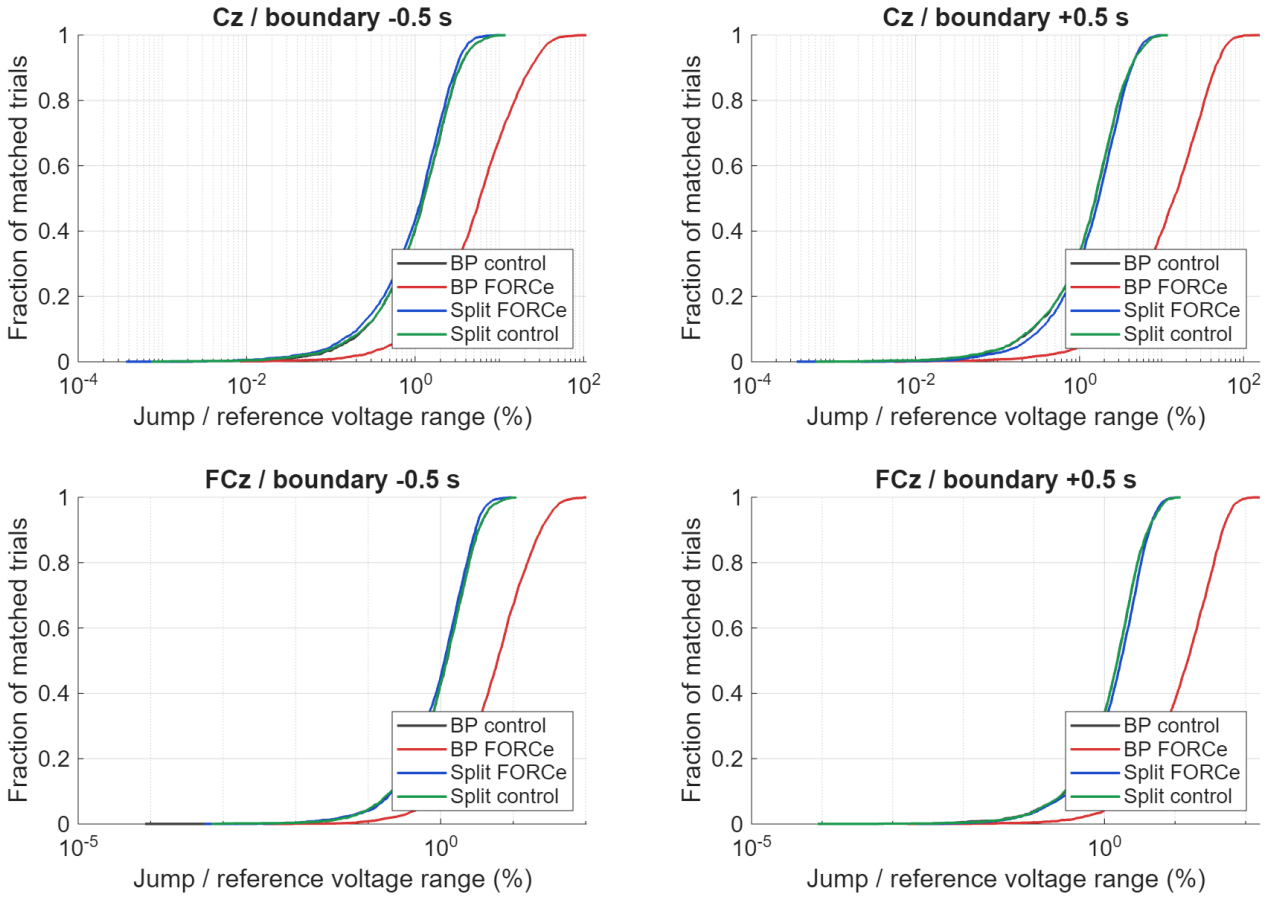


Figure 1: Matched-trial distributions of normalised jumps in Cz and FCz; BP denotes bandpass. The denominator is fixed across conditions. Onset and all-eight-channel results are supporting measurements.

3.3 Proposed output before and after low-pass

Ch.	Join	Median pre	Median post	Q_{95} pre	Q_{95} post	Max pre	Max post
Cz	-0.5	4.814	0.306	15.188	1.035	36.212	2.350
Cz	+0.5	6.511	0.440	20.112	1.445	39.980	2.787
FCz	-0.5	4.699	0.312	15.234	1.053	42.828	2.667
FCz	+0.5	6.805	0.473	21.413	1.554	47.801	2.973

All values are in μV and use the same proposed-cleaning trials before and after the final low-pass. Each maximum can arise from a different trial. The reduction demonstrates attenuation of adjacent-sample changes; it does not establish removal of broader effects.

3.4 Representative individual trials

Two examples are selected at the median and 95th-percentile rank of condition 2's Cz normalised jump at +0.5 s. Every overlay uses the same trial and channel. This selection illustrates the reference condition; it is not an average treatment effect.

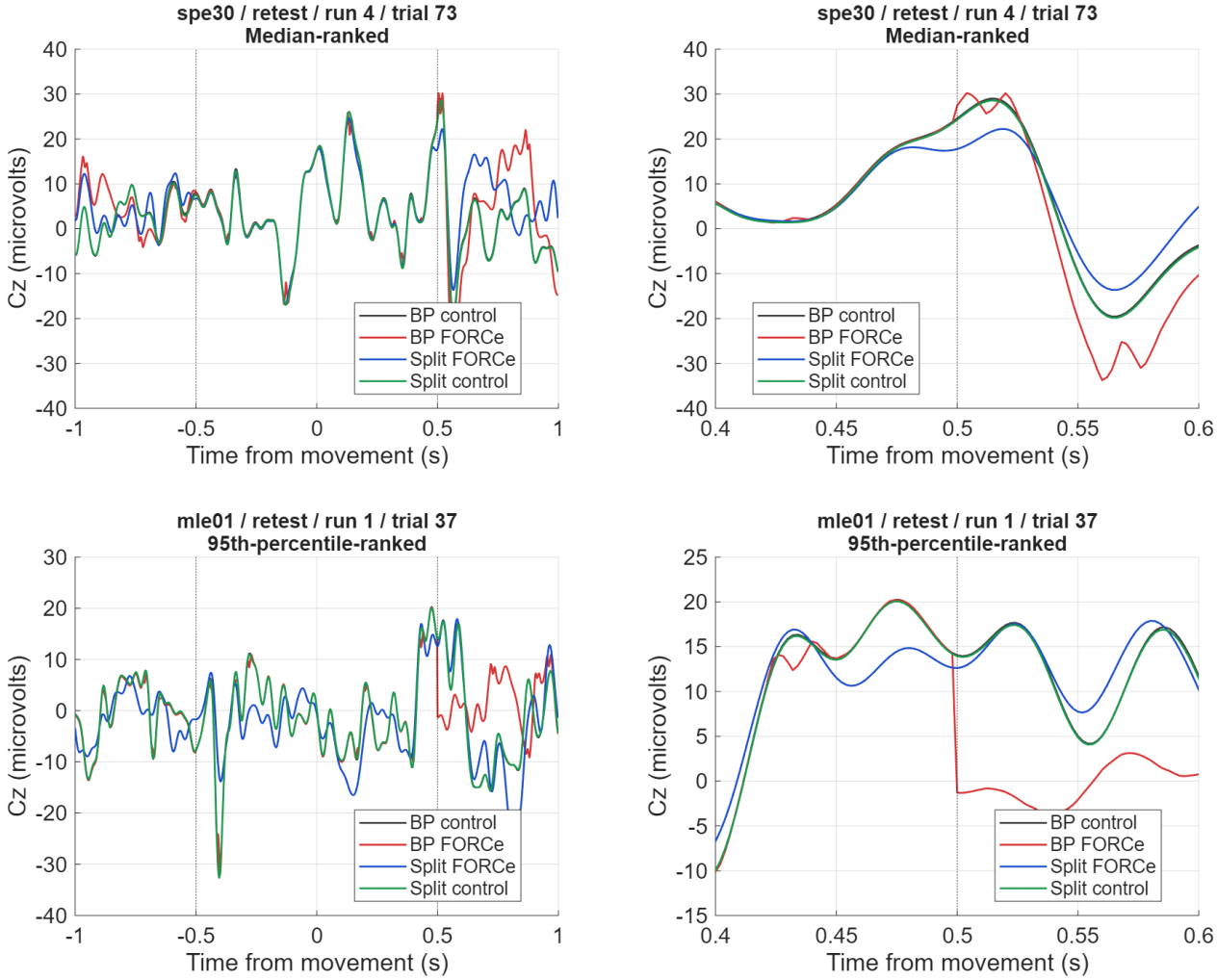


Figure 2: Full-epoch and join-detail overlays for representative trials. A smoother trace alone cannot establish preservation of the underlying EEG.

4 Results II: filtering and broader changes

4.1 Finite-segment filtering and synthetic boundaries

The largest finite-segment filtering residual in the matched Cz/FCz data is 6.28×10^{-10} μV anywhere in the retained epoch. This describes the tested extraction length and filter only.

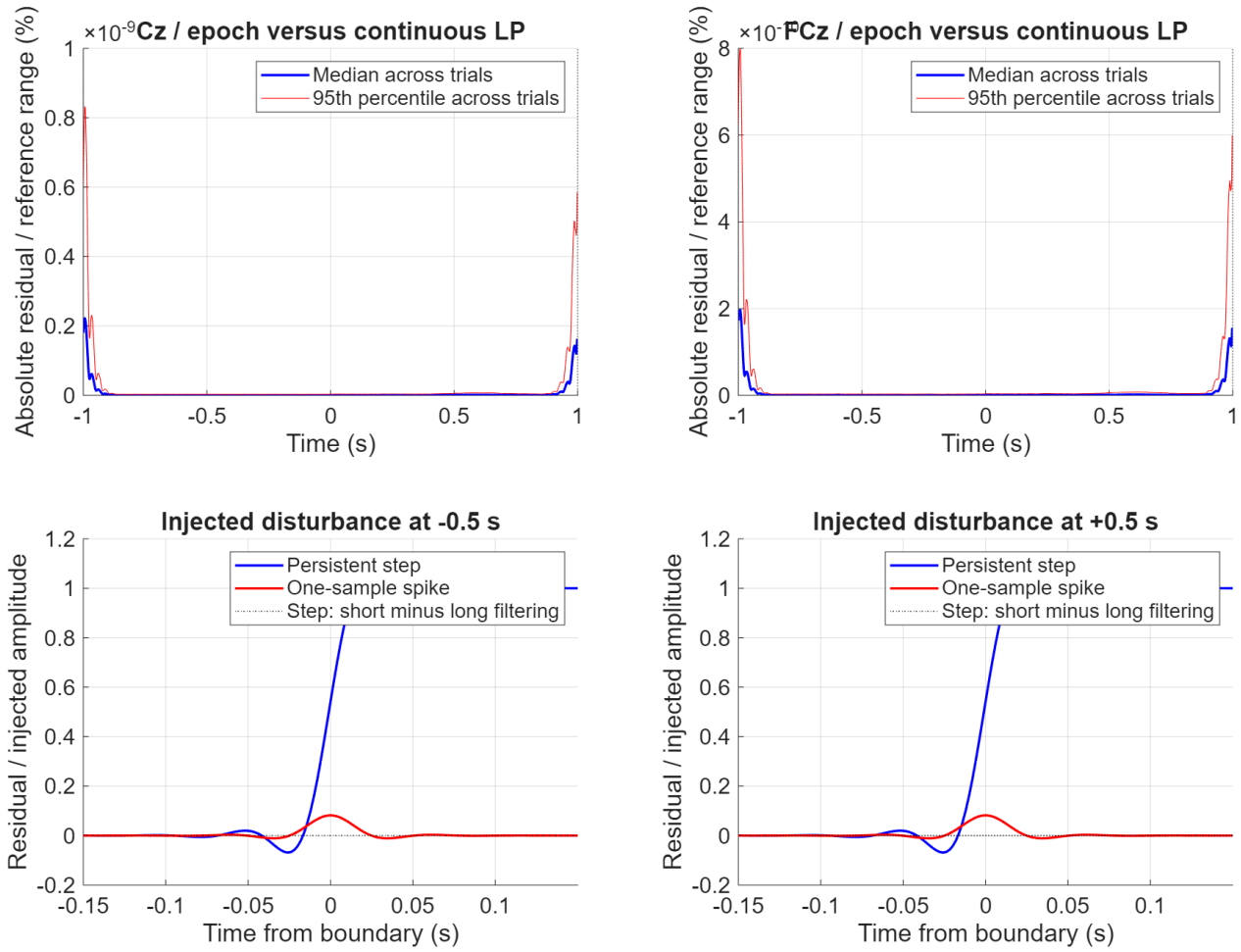


Figure 3: Top: absolute normalised filtering-only residual envelopes across matched trials. Bottom: persistent-step and one-sample-spike responses normalised by injected amplitude. A smoothed persistent step still retains a level change; short-minus-long filtering isolates endpoint effects.

4.2 Response-interval residuals

Contrast 1: finite-segment filtering. Contrast 2: split FORCe minus its control. Contrast 3: bandpass FORCe minus its control. Each entry is the median across trials of the stated within-trial statistic in 0.2–0.9 s; the final column is not a pooled percentile.

Contrast	Ch.	Median (μV)	Median/S (%)	Trial Q_{95}/S (%)
1	Cz	3.25×10^{-13}	1.21×10^{-12}	4.97×10^{-12}
1	FCz	3.36×10^{-13}	1.19×10^{-12}	5.00×10^{-12}
2	Cz	2.47350	9.502	49.343
2	FCz	2.80303	9.968	51.860
3	Cz	2.21061	8.298	44.863
3	FCz	2.48240	8.813	47.376

Cleaning residuals contain intended removal as well as possible unwanted changes. The appendix shows condition-difference waveforms and extrema. Changes in the largest peak can reflect a different selected extremum, rather than latency movement of the same physiological component.

4.3 Centring and reconstruction audit

Participant	Ch.	Windows	|shift| (μV)	Nonconstant RMS (μV)
mle01	Cz	135	0.601	0.000000
mle01	FCz	135	0.608	0.000000
spe30	Cz	126	0.838	0.000000
spe30	FCz	126	0.954	0.000000

Median absolute shift is 0.734 μV; median nonconstant RMS is 0.000000 μV. Nonconstant RMS values at floating-point resolution are not evidence of mathematically exact zero. The audit does not attribute all FORCe jumps to centring or justify restoring means without review.

5 Discussion and limits

The exact split-filter control distinguishes cleaning effects from filtering implementation effects. All four conditions are needed for this attribution even though condition 4 is not a candidate final pipeline. Three-second extraction provides 0.5 s of additional context beyond each cropped endpoint, but leaves joins at -0.5 and $+0.5$ s inside the retained epoch. A low-pass attenuates sharp changes; it need not remove a sustained shift. The synthetic response describes known disturbances, not every real FORCe reconstruction effect. Individual-trial distributions take priority over averages for the boundary question. Conversely, no-FORCe data are not neural ground truth, and a cleaning residual does not distinguish artifact from neural content. Visual smoothness, rarity, residual size and apparent peak changes must be interpreted together. The practical result is a strong reduction in sharp boundary steps under the proposed processing, with negligible outer-endpoint filtering effects for this setup. However, median response-interval cleaning residuals remain about 9.5–10.0% of the reference voltage range in the proposed condition. Those changes cannot be classified as artifact removal or neural distortion from this pilot alone. The pilot does not retune FORCe, benchmark artifact removal, establish classifier performance, or support population-level inference. Retaining FORCe should depend on useful artifact handling without unacceptable response changes. Proceeding without it remains an option.

A Participant-specific boundary statistics

Participant	Ch.	Join	ID	Median (μV)	J/S (%)	Rare (%)
mle01	Cz	-0.5	1	0.330	1.43	1.1
mle01	Cz	-0.5	2	1.426	5.87	39.5
mle01	Cz	-0.5	3	0.289	1.24	1.8
mle01	Cz	-0.5	4	0.328	1.42	1.1
mle01	Cz	+0.5	1	0.370	1.53	1.4
mle01	Cz	+0.5	2	2.816	11.93	60.6
mle01	Cz	+0.5	3	0.368	1.63	3.5
mle01	Cz	+0.5	4	0.369	1.51	1.4
mle01	FCz	-0.5	1	0.328	1.37	1.2
mle01	FCz	-0.5	2	1.421	5.90	40.1
mle01	FCz	-0.5	3	0.285	1.19	2.0
mle01	FCz	-0.5	4	0.325	1.36	1.2
mle01	FCz	+0.5	1	0.383	1.50	1.1
mle01	FCz	+0.5	2	3.049	12.43	62.7
mle01	FCz	+0.5	3	0.383	1.59	3.8
mle01	FCz	+0.5	4	0.385	1.51	1.1
spe30	Cz	-0.5	1	0.330	1.17	0.1
spe30	Cz	-0.5	2	1.686	5.81	44.7
spe30	Cz	-0.5	3	0.320	1.13	1.1
spe30	Cz	-0.5	4	0.327	1.15	0.1
spe30	Cz	+0.5	1	0.447	1.59	2.7
spe30	Cz	+0.5	2	4.730	16.83	72.2
spe30	Cz	+0.5	3	0.509	1.82	7.7
spe30	Cz	+0.5	4	0.452	1.57	2.9
spe30	FCz	-0.5	1	0.344	1.12	0.3
spe30	FCz	-0.5	2	1.928	6.24	46.8
spe30	FCz	-0.5	3	0.337	1.07	1.0
spe30	FCz	-0.5	4	0.344	1.11	0.2
spe30	FCz	+0.5	1	0.474	1.55	3.1
spe30	FCz	+0.5	2	5.495	18.03	73.8
spe30	FCz	+0.5	3	0.550	1.83	9.4
spe30	FCz	+0.5	4	0.476	1.54	3.1

B Onset and maximum-channel supporting summaries

Movement onset is not a join in these conditions. Maximum-channel measurements select each trial's largest raw step, then use that channel's ordinary-change distribution. They can select different channels across conditions.

Channel	Time	ID	Median (μV)	Q_{95} (μV)	Max (μV)	Rare (%)
Cz	+0.0	1	0.350	1.276	2.812	0.7
Cz	+0.0	2	0.363	1.316	3.273	1.1
Cz	+0.0	3	0.282	0.990	2.510	1.4
Cz	+0.0	4	0.344	1.262	2.806	0.7
FCz	+0.0	1	0.356	1.283	2.837	0.7
FCz	+0.0	2	0.377	1.340	3.304	1.1
FCz	+0.0	3	0.289	1.006	2.575	1.3
FCz	+0.0	4	0.352	1.279	2.833	0.7
Max-eight	-0.5	1	0.556	1.337	6.984	1.4
Max-eight	-0.5	2	2.242	11.481	83.707	57.7
Max-eight	-0.5	3	0.511	1.189	4.678	2.9
Max-eight	-0.5	4	0.550	1.329	6.958	1.3
Max-eight	+0.0	1	0.567	1.492	2.837	1.6
Max-eight	+0.0	2	0.646	1.527	3.306	2.9
Max-eight	+0.0	3	0.483	1.198	2.575	2.8
Max-eight	+0.0	4	0.564	1.480	2.833	1.6
Max-eight	+0.5	1	0.642	1.667	3.733	3.9
Max-eight	+0.5	2	5.262	19.637	39.407	81.8
Max-eight	+0.5	3	0.655	1.658	3.118	11.0
Max-eight	+0.5	4	0.637	1.647	3.752	3.8

C Condition-difference waveforms

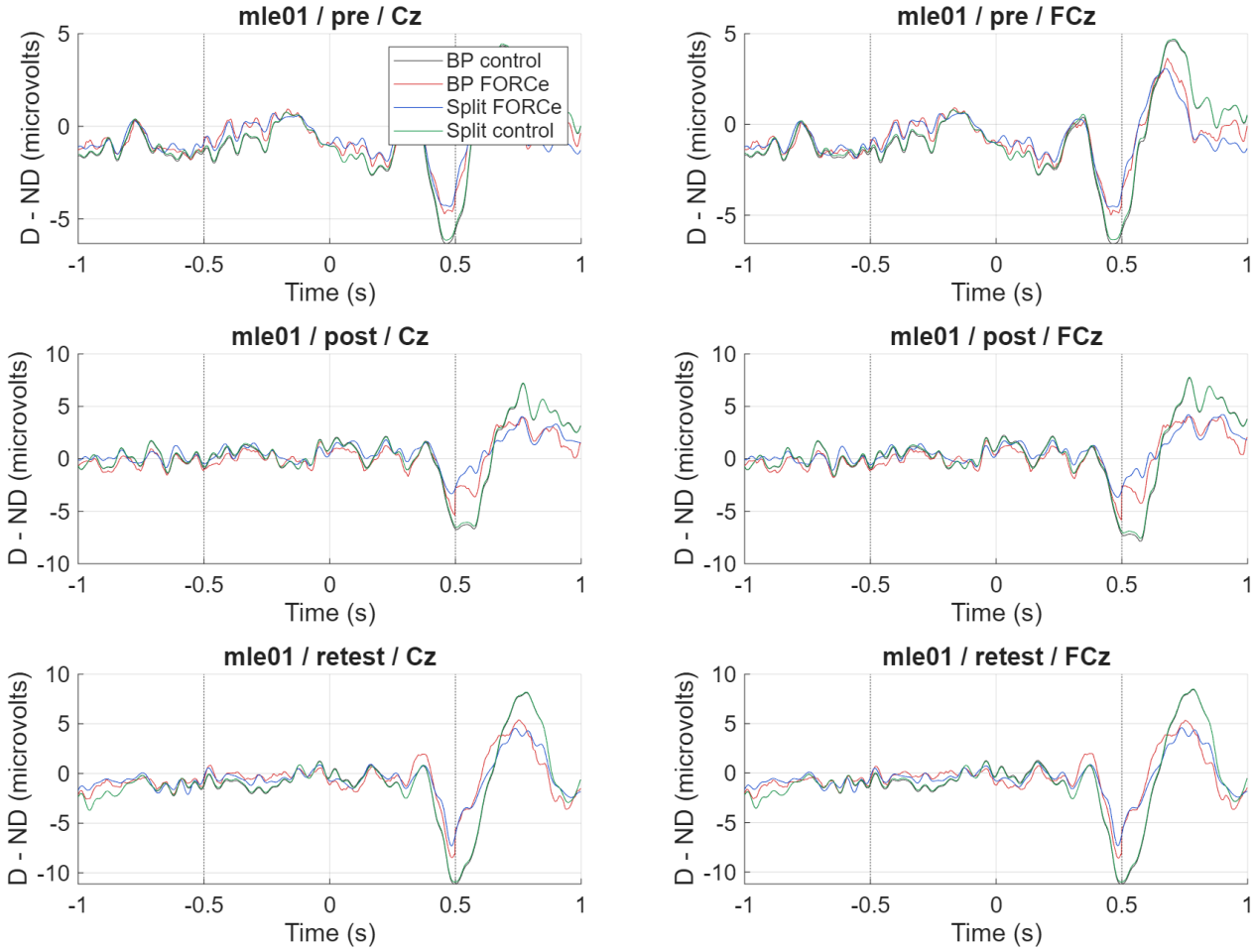


Figure 4: Matched displaced-minus-non-displaced waveforms for mle01, by session and channel. Joins are at -0.5 and +0.5 s.

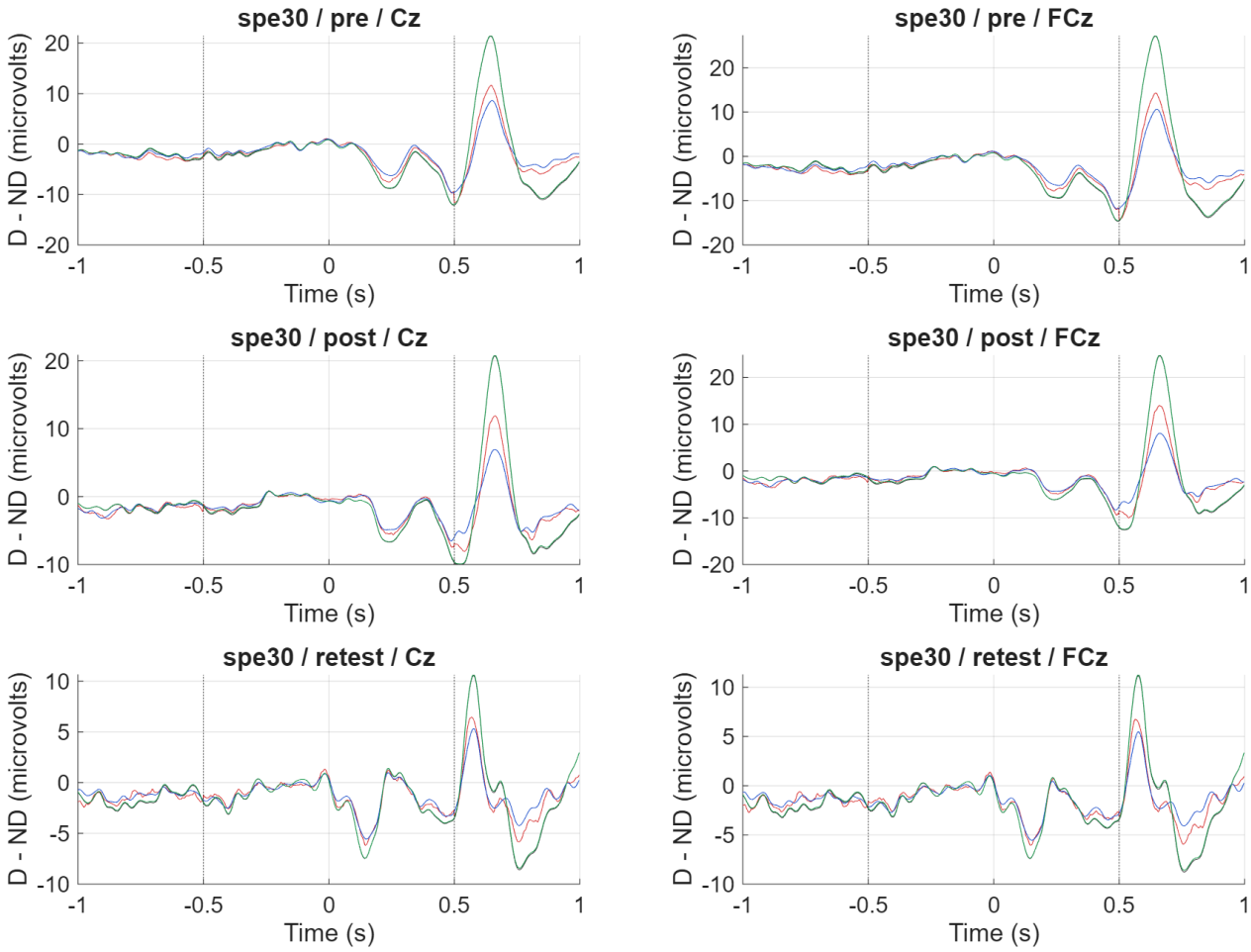


Figure 5: Matched displaced-minus-non-displaced waveforms for spe30, by session and channel. Joins are at -0.5 and +0.5 s.

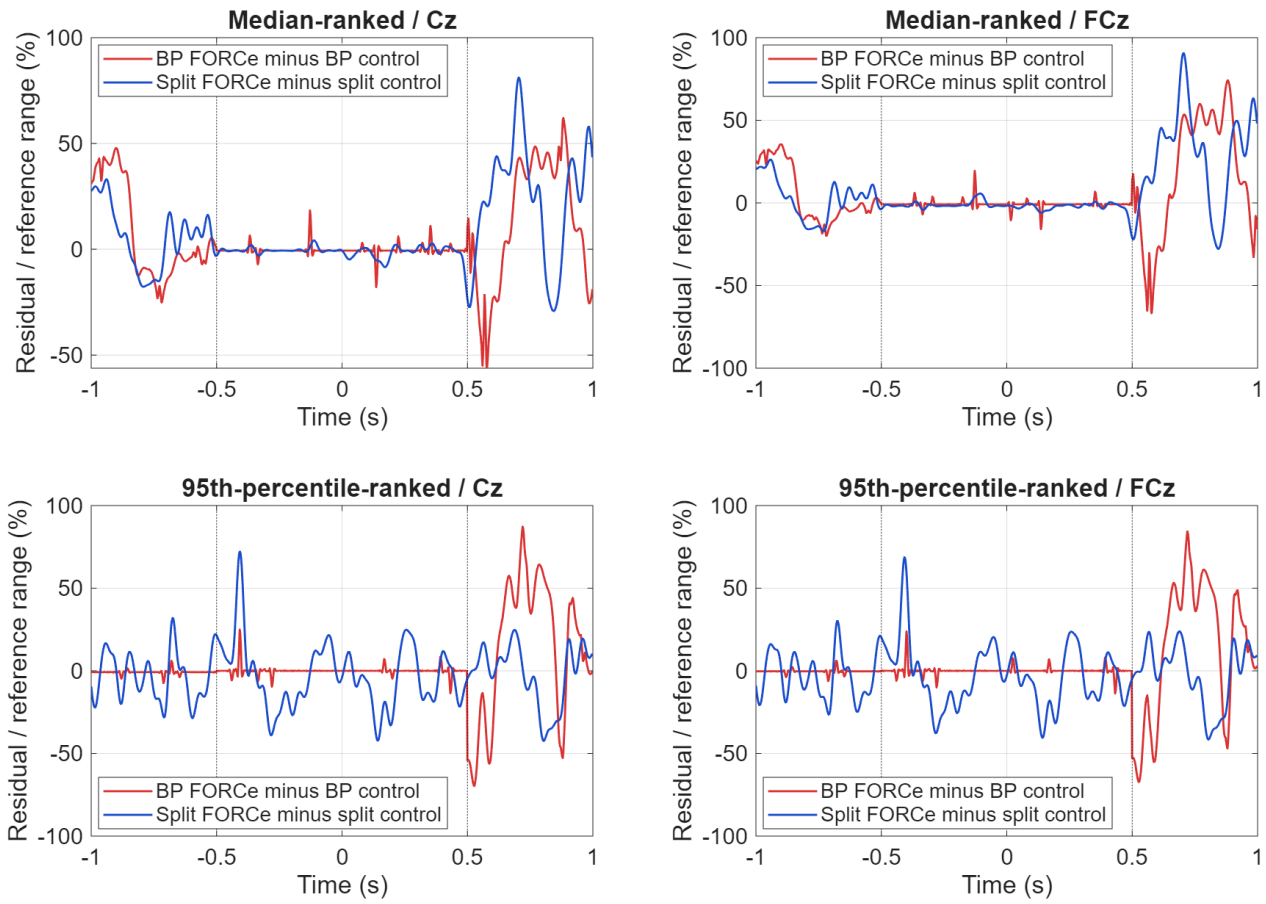


Figure 6: Actual cleaning residuals for the same representative trials, in Cz and FCz. Both curves use the common condition-1 voltage scale. These differences include artifact removal and are not neural-error estimates.

D ERP extrema and peak-time descriptions

Participant	Session	Ch.	ID	Min (μV)	Max (μV)	Peak time (s)
mle01	pre	Cz	1	-6.324	4.365	0.464
mle01	pre	Cz	2	-4.716	3.698	0.456
mle01	pre	Cz	3	-4.330	3.147	0.478
mle01	pre	Cz	4	-6.137	4.445	0.464
mle01	pre	FCz	1	-6.559	4.616	0.464
mle01	pre	FCz	2	-4.984	3.633	0.456
mle01	pre	FCz	3	-4.551	3.089	0.478
mle01	pre	FCz	4	-6.348	4.699	0.462
mle01	post	Cz	1	-6.746	7.167	0.770
mle01	post	Cz	2	-5.377	3.919	0.498
mle01	post	Cz	3	-3.334	4.027	0.766
mle01	post	Cz	4	-6.531	7.232	0.770
mle01	post	FCz	1	-7.878	7.694	0.574
mle01	post	FCz	2	-5.840	4.026	0.498
mle01	post	FCz	3	-3.659	4.214	0.898
mle01	post	FCz	4	-7.586	7.773	0.768
mle01	retest	Cz	1	-11.119	8.129	0.498
mle01	retest	Cz	2	-8.459	5.411	0.488
mle01	retest	Cz	3	-7.284	4.562	0.484
mle01	retest	Cz	4	-10.921	8.201	0.498
mle01	retest	FCz	1	-11.215	8.431	0.498
mle01	retest	FCz	2	-8.612	5.333	0.488
mle01	retest	FCz	3	-7.348	4.592	0.484
mle01	retest	FCz	4	-11.016	8.501	0.498
spe30	pre	Cz	1	-12.104	21.407	0.646
spe30	pre	Cz	2	-11.545	11.641	0.648
spe30	pre	Cz	3	-9.668	8.616	0.496
spe30	pre	Cz	4	-12.007	21.506	0.644
spe30	pre	FCz	1	-14.643	27.149	0.644
spe30	pre	FCz	2	-14.220	14.290	0.646
spe30	pre	FCz	3	-11.845	10.636	0.492
spe30	pre	FCz	4	-14.518	27.261	0.644
spe30	post	Cz	1	-9.988	20.703	0.662
spe30	post	Cz	2	-8.065	11.882	0.662
spe30	post	Cz	3	-6.511	6.950	0.662
spe30	post	Cz	4	-9.887	20.824	0.662
spe30	post	FCz	1	-12.577	24.655	0.662
spe30	post	FCz	2	-9.999	14.002	0.660
spe30	post	FCz	3	-8.319	8.076	0.486
spe30	post	FCz	4	-12.456	24.790	0.662
spe30	retest	Cz	1	-8.588	10.533	0.576
spe30	retest	Cz	2	-5.866	6.441	0.568
spe30	retest	Cz	3	-4.232	5.316	0.578
spe30	retest	Cz	4	-8.432	10.629	0.576
spe30	retest	FCz	1	-8.763	11.157	0.576
spe30	retest	FCz	2	-5.937	6.734	0.566
spe30	retest	FCz	3	-4.085	5.463	0.578
spe30	retest	FCz	4	-8.589	11.270	0.576

References

[1] Daly I, Scherer R, Billinger M, Müller-Putz G. FORCe: Fully Online and Automated Artifact Removal for Brain-Computer Interfacing. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*. 2015;23(5):725–736. doi:10.1109/TNSRE.2014.2346621.

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[3] Widmann A, Schröger E. Filter Effects and Filter Artifacts in the Analysis of Electrophysiological Data. *Frontiers in Psychology*. 2012;3:233. doi:10.3389/fpsyg.2012.00233.